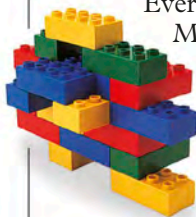


SOLIDS



Everything is made up of matter. Molecules are the building blocks of matter. In solids, molecules are usually packed together in a regular way, so the solid keeps its shape without needing a container to hold it. Most matter is visible, like

the pages of this book, but some matter is invisible, like air. Matter may be solid, liquid, or gas, and change from one state to another.



In a state

Ice, water, and steam are made up of the same molecules. They are not alike because their molecules are spaced differently. Solid ice melts into liquid water. When water is boiled, it turns into a gas called steam.

Hydrogen atom



Uranium atom



Amazing atoms

The smallest amount of a pure substance is called an atom. The simplest atom is hydrogen, but other atoms are more complex. Atoms are incredibly tiny—about 100,000 million atoms fit on this period.

Molecules in a solid

Each of these children is acting the part of a molecule in a solid.

The children are all wearing red T-shirts, to show that the molecules are all the same.

The molecules in most solids make up a very regular pattern.

Solids on show

You can tell these solids apart because they do not look, feel, weigh or smell the same. They all have different properties.



A flower is soft and delicate.



A cake is crumbly and tasty.